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Roll No

EE-601 (GS)

B.E. VI Semester Examination, June 2020

Grading System (GS)

Microprocessor and Microcontrollers

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Draw the pin diagram of 8086 microprocessor and explain function of each pin.

2. Explain the function of following instructions of 8086.
 i) XLAT ii) LOCK
 iii) PCHL iv) LES

3. Write a short notes on the following:
 i) Subroutine call and return
 ii) Stack structure of 8086

4. List some applications of microprocessors. With a neat diagram, explain the internal architecture of the 8086 microprocessor.

5. Explain the interfacing of ADC with 8051 using 8255 and write an assembly language program to converter and read equivalent digital data from ADC.

6. Write a short note on any two of the following:
 a) Stepper motor interfacing with 8051 μC
 b) 8051 based thyristor firing circuit
 c) Register indirect addressing mode

7. Differentiate between microcontroller and microprocessor. With a neat diagram, explain the internal architecture of 8051 microcontroller. Also list different applications of microcontrollers.

 OR

 Why stack is Last In First Out (LIFO) data segment? With the help of suitable diagram explain stack-top address calculation.

8. Answer any two of the following:
 a) Discuss the salient feature of 80286 as far as memory addressing is concerned.
 b) Write an assembly language program for 8086 for Multiplication of two numbers.
 c) Draw the architecture of DMA controller chip.
 d) Write short note on the instruction set of 8051.

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